

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

March 2021

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

Question Paper No. : U033

Duration: 3 hours

Max. Marks: 100

PART - A

10 x 2 = 20 Marks

Answer all the questions

1. Expand $\tan 5\theta$ in powers of $\tan \theta$.
2. Find the general value of $\text{Log}(1+i) + \text{Log}(1-i)$.
3. Show that the vectors $(1, 2, 2)$, $(2, 1, -2)$, $(2, -2, 1)$ are linearly independent.

4. Two eigenvalues of the matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ are equal to 1 each.

Find the eigenvalues of A^{-1} .

5. Find the nature of the quadratic form $8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - 8x_2x_3 + 4x_3x_1$.
6. Find the sum of squares of the roots of $x^4 - 12x^3 + 47x^2 - 72x + 36 = 0$.
7. Find the coefficient of x^n in the expansion of $\frac{1+x+x^2}{(1-x)^2}$.

8. Find k if $\left(1 + \frac{1}{2!} + \frac{1}{4!} + \dots\right)^2 = k + \left(1 + \frac{1}{3!} + \frac{1}{5!} + \dots\right)^2$.

9. Evaluate $\int_0^{\pi/2} \sin^5 \theta d\theta$

10. Prove that $\int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx = \Gamma(n)$

PART- B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Show that $\frac{\sin 7\theta}{\sin \theta} = 7 - 56 \sin^2 \theta + 112 \sin^4 \theta - 64 \sin^6 \theta$ (7)

(b) If $\cos(x + iy) = r(\cos \alpha + i \sin \alpha)$, prove that

(i) $2r^2 = \cos 2x + \cosh 2y$

(ii) $y = \frac{1}{2} \log \left[\frac{\sin(x - \alpha)}{\sin(x + \alpha)} \right]$ (8)

(OR)

12. (a) If $\tan(A + iB) = x + iy$, prove that

(i) $x^2 + y^2 + 2x \cot 2A = 1$ (7)

(ii) $x^2 + y^2 + 1 - 2y \coth 2B = 0$

(b) Expand $\cos^5 \theta \sin^4 \theta$ in series of cosines of multiples of θ . (8)

13. (a) Test for the consistency of the system of equations

$$x + 2y - 3z = 3$$

$$2x + y + 4z = 1$$

$$5x + 4y + 4z = 2$$

and find the solution (if it exists).

(5)

(b) Find the eigenvalues and eigenvectors of the matrix

$$\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix} \text{ and hence diagonalise it.} \quad (10)$$

(OR)

14. (a) Verify Cayley Hamilton theorem for the matrix $\begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$

and hence find A^{-1} and A^4 . (10)

(b) The eigenvectors of a 3×3 real symmetric matrix A corresponding to the eigenvalues 2, 3, 6 are

$(1, 0, -1)^T, (1, 1, 1)^T, (-1, 2, -1)^T$ respectively. Find A . (5)

15. (a) Find the condition that the roots of the equation $x^4 + px^3 + qx^2 + rx + s = 0$ may be such that the product of two roots is equal to the negative of the product of the other two roots. (7)

(b) Find the sum to infinity of the series

$$\frac{5}{3.6} + \frac{5.7}{3.6.9} + \frac{5.7.9}{3.6.9.12} + \dots \quad (8)$$

(OR)

16. (a) Diminish the roots of $x^4 - 18x^3 + 110x^2 - 266x + 221 = 0$ by 2 and hence solve the equation. (7)

(b) Sum the series:

$$1 + \left(\frac{1}{2} + \frac{1}{3}\right)\frac{1}{9} + \left(\frac{1}{4} + \frac{1}{5}\right)\frac{1}{9^2} + \left(\frac{1}{6} + \frac{1}{7}\right)\frac{1}{9^3} + \dots \quad (8)$$

17. (a) Prove that $\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$ (7)

(b) Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy dy dx$ and hence evaluate. (8)

(OR)

18. (a) Prove that $\int_0^1 x^m (1-x^p)^n dx = \frac{1}{p} \beta\left(\frac{m+1}{p}, n+1\right)$ and hence find

$\int_0^1 x^4 (1-x^2)^3 dx$ (7)

(b) Evaluate $\int_0^{\log_e 2} \int_0^{x+y} \int_0^z e^{x+y+z} dx dy dz$ (8)

PART -C

Answer the following

1 x 20 = 20 Marks

19. By reducing the quadratic form

$3x_1^2 - 3x_2^2 - 5x_3^2 - 2x_1x_2 - 6x_2x_3 - 6x_3x_1$ to canonical form, determine its rank, index, signature and nature of the quadratic form.
